Team 20: Generation for Recommendation

Tony Zhao sz3822@nyu.edu
Jerry Huang jh8186@nyu.edu
Conny Fan jf4644@nyu.edu

Milestone 1

1. Motivation

Traditional music recommendation systems primarily focus on retrieving existing tracks that match a user's historical listening patterns. In contrast, this project explores a generative alternative that synthesizes new music conditioned on a user's short-term preference embedding. By leveraging audio embeddings, retrieval-based metadata abstraction, and large language models for prompt construction, we investigate whether generative personalization can produce musically coherent outputs that align with a user's dominant listening preferences.

Recent advances in generative audio models have made it possible to move beyond item retrieval toward active content creation. However, an important challenge remains: how to translate a user's historical listening patterns into controllable generative signals while preserving stylistic coherence. In particular, representing user preferences within an audio embedding space introduces the question of how such representations can effectively guide the generation process.

This project therefore studies how user listening history can be transformed into conditioning signals for personalized AI music generation. We specifically investigate whether coherence-filtered, temporally weighted user embeddings provide a more accurate representation of short-term listening preferences than naïve embedding averaging, and whether retrieval-based metadata abstraction improves stylistic alignment between generated music and user preferences compared to direct prompt-based generation.

2. Related Work

2.1 Related Work 1

Plitsis et al. (2023) studies few-shot personalization of text-to-audio music diffusion models. Starting from a pretrained text-to-audio latent diffusion model (AudioLDM), their model learns a new musical concept from only a few audio clips, while keeping the concept editable via textual prompts. The study compares two established approaches: Textual Inversion (optimize only the new token embedding) and DreamBooth (fine-tune the denoising network). Furthermore, evaluation is through reconstruction and editability, using a newly collected dataset with 32 concepts $\times$ 5 clips $\times$ 10s, plus 20 editability prompts. It uses CLAP-A / CLAP-T embeddings derived from the CLAP model (Elizalde et al. (2022)) and FAD, plus music-specific metrics like BPM, loudness in LUFS, and

key/scale via HPCP (computed with Essentia). In addition, a human A/B user study is performed to assess perceptual quality.

A strength of this work is that the evaluation is multi-view rather than relying on a single metric, which is relevant to our concern about metric bias when an embedding space is used as both representation and judge. We also learned that current personalization approaches learn rhythmic constructs more easily than melody, so we will keep this in mind in our later work.

A limitation is that the few-shot personalization is evaluated on a relatively small set of concepts (32), with only five 10-second clips per concept and 20 editability prompts. As a result, it mainly supports concept-level adaptation rather than personalization driven by long-term user listening histories like our pipeline.

2.2 Related Work 2

Ma et al. (2022) presents a closed-loop personalization pipeline: it learns a User Music Preference (UMP) embedding from listening history, computes the distance between a seed song and the UMP, converts that distance into interpretable control parameters, and generates a new song that moves closer to the user preference in embedding space. It proposes a hierarchical UMP embedding model (including CRNN and self-attention) that maps music features into song embeddings, and defines a user's UMP as a weighted average over embeddings of listened songs. It then uses an embedding-distance model to map the seed-vs-UMP gap into a music-distance hyperspace with four axes (chord progression, melody statistics, melodic rhythm, and melodic contour), which are fed into a controllable generator to produce preference-aligned music.

A strength is that the overall loop is conceptually close to our project: the embedding process represents the user's preference as a vector, converts distance-to-preference into generation-time control signals, and iteratively steers generations toward the user.

A limitation is that the work's approach differs from our planned pipeline in important ways. In particular, their method relies on a symbolic, parameterized generation setup driven by a small set of control parameters, whereas we aim to build user embeddings in an audio semantic space with CLAP and implement personalization through a modular system: structured profile → prompt → multi-candidate generation → CLAP-based reranking. Therefore, we refer to the logic and the trade-offs discussed in this work, while adopting a different implementation path.

3. Methodology

We structure our system into five phases: (1) Song Embeddings, (2) User (Weighted) Embeddings, (3) User Profile & Prompt Generation, (4) Music Generation, and (5) Evaluation.

3.1 Phase 1: Song Embeddings

We first embed each Music4All track using a pre-trained CLAP model into a shared 512-dimensional representation space, where cosine similarity provides a direct measure of song-to-song relatedness. We then perform contrastive fine-tuning of CLAP on Music4All audio–text/tag pairs using a PyTorch training pipeline (audio loading via `torchaudio`, text/token processing via a tokenizer, batched `DataLoader`, and gradient-based optimization). The contrastive objective pulls matched audio–text pairs closer together while pushing mismatched pairs apart, improving semantic alignment between audio and metadata within the CLAP embedding space.

3.2 Phase 2: User (Weighted) Embeddings

We represent each user as a 512-dimensional vector by aggregating CLAP embeddings of their Recent-K listened songs with a temporal-decay weighting scheme, so more recent and/or more frequently played tracks receive higher weight. This yields a short-term preference representation in the same semantic space as song embeddings, enabling user-to-user and user-to-song similarity comparisons via cosine similarity.

Recent listening histories can be diverse, and a naive weighted mean over all Recent-K tracks can produce an "in-between" embedding that does not correspond to any coherent musical style. To mitigate this, we introduce a coherence check. We compute pairwise cosine similarities among the Recent-K song embeddings, select a representative medoid track (the track with the highest average similarity to the others), and discard tracks whose similarity to the medoid falls below a threshold as outliers. We then compute the temporally-decayed weighted user embedding over this most cohesive subset. This produces a more coherent user representation that reflects the dominant short-term listening style rather than averaging across unrelated musical modes.

3.3 Phase 3: User Profile & Prompt Generation

To convert user embeddings into interpretable preferences and controllable generation prompts, we retrieve the Top-M songs closest to the user embedding in CLAP space. Using song IDs, we join Music4All metadata (e.g., genres, tags, and other attributes) and aggregate these fields into a structured user profile (e.g., weighted tag/genre frequencies and dominant mood/instrument/tempo cues). This aggregation captures the user's dominant short-term taste while reducing noise from idiosyncratic listens.

Finally, we pass the structured profile to an LLM to produce (i) a concise natural-language preference summary and (ii) a standardized prompt template that will condition the music generation model. The LLM serves as a semantic abstraction layer that converts structured metadata statistics into a coherent natural-language description suitable for conditioning a generative music model.

3.4 Phase 4: Music Generation

We generate personalized audio conditioned on the user-specific prompt produced in Phase 3. The generator model is selected based on feasibility and domain fit. If we determine that adaptation is necessary to better match the Music4All domain or target

styles, we will prioritize an open-source generator (e.g., MusicGen) to enable direct fine-tuning or lightweight parameter-efficient tuning. If fine-tuning is not required, we will use a strong pre-trained music generation model (e.g., Lyria 3) and focus on prompt conditioning.

For each user prompt, we sample multiple candidate clips. We embed each generated clip using CLAP and re-rank candidates by cosine similarity to the user embedding, selecting the highest-scoring outputs as final results. Optionally, we enforce basic diversity constraints to avoid near-duplicate generations among the top-ranked candidates.

3.5 Phase 5: Evaluation

We use human judgment as the primary measure of quality and personalization. We conduct a user study where participants listen to generated clips alongside reference tracks representing the user's Recent-K/Top-M profile. Raters evaluate each clip on (1) preference alignment (does it sound like what this user would listen to), (2) musical quality/coherence, and (3) novelty (not a trivial copy of a known song).

We also include pairwise comparisons between methods (e.g., our full approach vs. a baseline prompt strategy or a non-personalized generator), asking raters to choose which output better matches the user profile.

4. Dataset

We utilize *Music4All* as the primary dataset for this project. Music4All is a large-scale music database containing rich multimodal information, including metadata, tags, genre annotations, audio clips, lyrics, and user listening records. This dataset is uniquely suitable for personalized music generation because it provides the ground-truth connectivity needed to link user preferences with generative model outputs.

First, the combination of audio clips and fine-grained tags (e.g., mood, instrument, and style) enables domain-specific adaptation of our audio–text embedding model. In particular, we can fine-tune CLAP on Music4All audio–tag/text pairs so that the resulting latent semantic space better reflects real-world musical attributes and nuanced similarity relationships within this corpus.

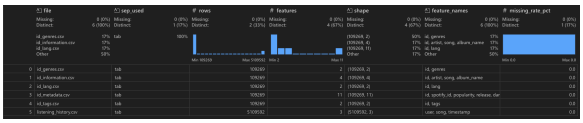

Figure 1: Dataset Overview

Second, the `listening_history.csv` file is critical for our User Embedding phase: it provides behavior-level evidence of what users actually listen to over time, allowing us to model *Recent-K* listening patterns rather than relying on static or self-reported profiles. This temporal signal is essential for capturing short-term preference shifts and producing prompts that reflect a user's current taste. In short, music4all integrates both song metadata and user behaviors, allowing us to set a foundation for further modeling and generation steps.

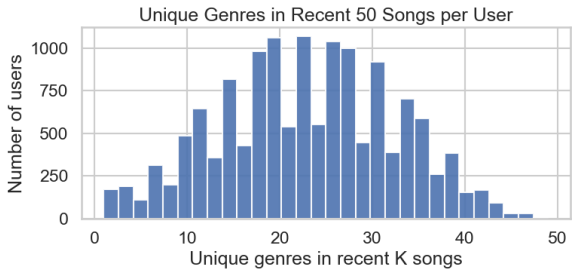

Figure 2: Unique Genres in Recent-K Songs per User

5. Workplan

5.1 Contribution so far:

From Week 1, our group started to form project ideas. Each member formed one idea with a detailed description, related work, and methodology; after discussion, we thought that Jerry's music generation is comparably more novel and fun for us. Plus, we have one member(also Jerry) in the band, and had used the CLAP model before.

Once we finalized the topic, we started to do heavy research on the pipeline, more models to use, and validation methods separately, and wrote down all we found and suggestions in a shared document(week 2-5). We also did 2 meetings with the assigned TA throughout the weeks. This week, we separate the parts for milestone 1(Tony: Motivation & Methodology & Dataset, Conny: Related Works & Work Plan, Jerry: Methodology & Dataset & Communication with TA), and take responsibility to work on each's parts and compile the materials into overleaf.

In the meantime, we communicate in our group chat on a daily basis, throw up ideas, and schedule member-only group meetings before each class.

5.2 Tentative Weekly Plan:

Week 1 - 6:

- Project Outline & Data Understanding
- Dataset EDA
- Pipeline Skeleton
- Milestone 1 deliverable

Week 7:

- Extract CLAP embeddings for Music4All tracks (batch processing + caching)
- Implement user embeddings from Recent-K with temporal decay weighting

- Implement coherence filtering (medoid selection + outlier removal)
- Sanity checks: nearest-neighbor retrieval for a small set of users/tracks

Week 8: Spring Break

Week 9:

- Retrieval + profile construction (Top-M reference tracks + metadata aggregation)
- Prompt forming (start with a deterministic template; LLM-based rewrite optional)
- First-round music generation with an initial generator choice (MusicGen / Lyria 3)
- Multi-candidate sampling (Q candidates per prompt) + CLAP embedding + reranking
- Milestone 2 deliverable

Week 10:

- Define baselines
- Run a small user study and refine instructions/rubrics
- Determine generator choice(if possible)

Week 11:

- If needed, finetune CLAP
- Contrast finetune CLAP on Music4All
- Ablation: original CLAP vs fine-tuned CLAP for reranking (same candidate pool)
- Report whether fine-tuning improves human-rated alignment and/or objective metrics

Week 12:

- Prompt engineering: template variants, negative constraints, length control
- Diversity / de-duplication strategy (similarity threshold or MMR)
- Finalize generator decision (only if not determined in Week 10)

Week 13:

- Full-scale evaluation on whole dataset and more users
- Analyze failure cases and identify what drives mismatches

Week 14:

- Milestone 3 Deliverable
- Prepare slides and presentation

References

Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang. Clap: Learning audio concepts from natural language supervision, 2022. URL https://arxiv.org/abs/2206.04769.

Xichu Ma, Yuchen Wang, and Ye Wang. Content based user preference modeling in music generation. In *Proceedings of the 30th ACM International Conference on Multimedia*, MM '22, page 2473–2482, New York, NY, USA, 2022. Association for Computing Machinery. ISBN 9781450392037. doi: 10.1145/3503161.3548169. URL https://doi.org/10.1145/3503161.3548169.

Manos Plitsis, Theodoros Kouzelis, Georgios Paraskevopoulos, Vassilis Katsouros, and Yannis Panagakis. Investigating personalization methods in text to music generation, 2023. URL https://arxiv.org/abs/2309.11140.